

Multiple Choice

1. **Answer:** D

Options: A) It relates inputs to outputs.; B) It is used for prediction.; C) It may be used for interpretation.; D) It discovers causal relationships

Note: The statement is false because regression analysis identifies correlations between variables but does not establish causation, as it cannot account for confounding factors or the direction of influence.

2. **Answer:** B

Options: A) The use of sampling with replacement as the sampling technique; B) The use of weak classifiers; C) The use of classification algorithms which are not prone to overfitting; D) The practice of validation performed on every classifier trained

Note: The use of weak classifiers in bagging reduces overfitting by averaging their predictions, which helps to mitigate the variance that can arise from complex models while maintaining a strong overall predictive performance.

3. **Answer:** A

Options: A) anomaly detection; B) one-class detection; C) train-test mismatch robustness; D) background detection

Note: Anomaly detection refers to the identification of data points that deviate significantly from the norm, which aligns with the concept of out-of-distribution detection, where the goal is to recognize instances that differ from the training data distribution.

4. **Answer:** B

Options: A) Increase the amount of training data.; B) Improve the optimisation algorithm being used for error minimisation.; C) Decrease the model complexity.; D) Reduce the noise in the training data.

Note: Improving the optimization algorithm primarily affects the convergence speed and efficiency of finding a minimum, but does not directly address the complexity of the model or its tendency to fit noise in the training data, which are the main causes of overfitting.

5. **Answer:** B

Options: A) $O(1)$; B) $O(N)$; C) $O(\log N)$; D) $O(N^2)$

Note: The correct answer is $O(N)$ because the nearest neighbors algorithm requires comparing the test instance to each of the N instances in the training dataset to determine the closest neighbors for classification.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: Linear Soft margin SVM allows for the classification of data that is not perfectly linearly separable by introducing a penalty for misclassification, enabling it to handle overlaps between classes.

7. **Answer:** True

Options: A) True; B) False

Note: Bayesians incorporate prior distributions to express beliefs about parameters before observing data, while frequentists rely on the data alone for inference, often avoiding the use of priors.

8. **Answer:** True

Options: A) True; B) False

Note: RoBERTa enhances BERT's training methodology by utilizing a significantly larger dataset, specifically around 160 GB of text data compared to BERT's 16 GB, which allows it to learn more robust language representations.

9. **Answer:** False

Options: A) True; B) False

Note: The rank of the matrix A is 1 because all its rows are identical and therefore linearly dependent, indicating that it can be reduced to a single row of non-zero values.

10. **Answer:** True

Options: A) True; B) False

Note: Neural networks can combine different activation functions in their architecture to leverage the strengths of each function for various layers, enhancing model performance and flexibility in learning complex patterns.

Long Form Response

11. **Answer:** `def find_last_occurrence(A, x): | return result`

Note: correct because it outlines the essential structure of a function that should implement a search algorithm, such as binary search, to efficiently locate the index of the last occurrence of a specified number in a sorted array.

12. **Answer:** `def Find_Max_Length(lst): | return maxLength`

Note: correct because it defines a function that aims to identify the maximum length of contiguous sublists within the given list, employing the concept of utilizing Python's built-in functions to effectively compute the desired outcome.

End of Answer Key